

13/08/26

Signal Processing

- Recap: Discrete-time signals
- Discrete-time systems
 - Memoryless, linear, time-invariant, causal, stable
 - Linear time-invariant systems
 - Examples: Delay, average, square, log, accumulator, compressor, forward and backward difference
- A discrete-time system $T\{\}$ is a function or a map from the input sequence $x[n]$ to a desired output sequence $y[n]$.

$$\begin{array}{ccc} & y[n] = T\{x[n]\} & \\ \downarrow & & \downarrow \\ \text{output} & & \text{input} \\ \text{sequence} & & \text{sequence} \end{array}$$

- We will carry out a systematic study of systems which satisfy various properties. These include: memoryless, linear, time-invariant, causal, stable, systems.
- Memoryless system: A system $T\{\}$ is said to be memoryless if the output at time index n depends only on the input at the same time index n (and not on any other time indices).

Ex 1: $y[n] = x[n]$ ✓

Ex 2: $y[n] = (x[n])^2$ ✓

Ex 3: $y[n] = \log(1 + x[n])$ ✓

Ex 4: $y[n] = \frac{x[n] + x[n-1]}{2}$ ✗

Ex 5: $y[n] = k$ for all n ?

• Linear system: A system $T\{\}$ is said to be linear if it satisfies the additivity and scaling properties.

- Additivity: A system $T\{\}$ is additive if it satisfies the following property: for any $x_1[n]$, $x_2[n]$, with $y_1[n] = T\{x_1[n]\}$,

$$y_2[n] = T\{x_2[n]\},$$

$$T\{x_1[n] + x_2[n]\} = T\{x_1[n]\} + T\{x_2[n]\}.$$

i.e., addition of the inputs implies addition of corresponding outputs

Ex 1: $y[n] = (x[n])^2$ is not linear

$$\text{If } y_1[n] = T\{x_1[n]\}, y_1[n] = (x_1[n])^2. \text{ || } y_2[n] = T\{x_2[n]\} = (x_2[n])^2$$

$$T\{x_1[n] + x_2[n]\} = (x_1[n] + x_2[n])^2 \neq y_1[n] + y_2[n]$$

- Scaling: A system $T\{\}$ is said to satisfy the scaling property if input scaling leads to corresponding scaling in the output i.e.,

$$\text{if } y[n] = T\{x[n]\}, T\{\alpha x[n]\} = \alpha \cdot T\{x[n]\} = \alpha \cdot y[n]. (\alpha \in \mathbb{R})$$

$$\text{Ex 1: If } y[n] = T\{x[n]\} = (x[n])^2, T\{\alpha x[n]\} = \alpha^2 (x[n])^2 = \alpha^2 \cdot y[n]$$

$$\alpha \cdot T\{x[n]\} = \alpha \cdot y[n].$$

$$\therefore T\{\alpha x[n]\} \neq \alpha \cdot T\{x[n]\}.$$

Putting these together compactly, a system $T\{\}$ is linear if $y_1[n] = T\{x_1[n]\}$, $y_2[n] = T\{x_2[n]\}$ and for arbitrary α, β , the following condition holds (for all $x_1[n], x_2[n]$)

$$T\{\alpha x_1[n] + \beta x_2[n]\} = \alpha \cdot T\{x_1[n]\} + \beta \cdot T\{x_2[n]\}$$

Ex 1: Let $y[n] = T\{x[n]\} = x[n-n_0]$ for some $n_0 > 0$. (Delay system)

$$\text{Let } x_3[n] = \alpha x_1[n] + \beta x_2[n]$$

$$\Rightarrow T\{x_3[n]\} = x_3[n-n_0]$$

$$= \alpha \cdot x_1[n-n_0] + \beta x_2[n-n_0]$$

(simply from defn of $x_3[n]$)

$$= \alpha \cdot T\{x_1[n]\} + \beta \cdot T\{x_2[n]\}$$

(from defn of $T\{\cdot\}$)

$$= \alpha \cdot y_1[n] + \beta y_2[n].$$

$\therefore$ the delay system is linear.

We can generalize the definition as

$$T\left\{\sum_{k=1}^K \alpha_k x_k[n]\right\} = \sum_{k=1}^K \alpha_k T\{x_k[n]\}$$